The unreasonable effectiveness of volatility targeting - and where it falls short

Part 1: Introduction, a paradox & blindspots

MARK ARON SZULYOVSZKY

APR 22, 2025

27 4 2 Sh

This is part 1 of our in-depth investigation of how quantitative risk management co help improve risk-adjusted returns: I'll explain what volatility targeting is, explore a seemingly paradoxical phenomenon, and highlight its blindspots.

Volatility targeting's goal is to keep an asset or portfolio's volatility within reasonabl boundaries, by actively adjusting exposure - it's among the very few *effective, universal* techniques in quantitative finance. I can't really think of many other than "diversification" and volatility targeting, if I had to answer on the spot.

A quick intro: *inverse volatility scaling* means when an asset is **getting more volati** you should **reduce** your exposure to it. People usually measure volatility with a rolli standard deviation of an asset's returns. It looks usually something like this:

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

Manage Reject Accept

Volatility usually *clusters* — when the markets are volatile, they stay volatile for som time. You can kind of see it on the chart above, it doesn't go "go back immediately" its starting position [1]. But, it does mean revert after some time.

Volatility targeting is using inverse volatility scaling with a pre-defined (annualized) volatility target leve Bitcoin's is around 60%.

You can make your by using the same volatility targ contributio each asset within a t.

What's the o

I'd say "*buying the* scaling car translated into: "*re*

And buying the dip stematic strategies that trac are *quantitative equiva* d as such, fo transparency reaso effectivene

If you plot the average 30 day forward returns when the SP500 was in a drawdown, you'd see this [2]:

Discover more from Unexpected Correlations by Unravel

Discover the predictive factors shaping market movements. In-depth reports, backtests and deployable code. By Unravel (unravel.finance): Market-netural, cros...

Enter your email...

Subscribe

By subscribing, you agree Substack's Terms of Use, and acknowledge its Information Collection Notice and Privacy Policy.

Already have an account? Sign in

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

The higher the drawdown, the larger the 30 day forward returns (SPY). Check footnote for the caveat with these plots.

This chart indicates that you should have bought the most when SP500 is in the biggest drawdown — thank you for that, captain hindsight! Of course this is not something that can be effectively executed, and definitely can not be called a risk management technique.

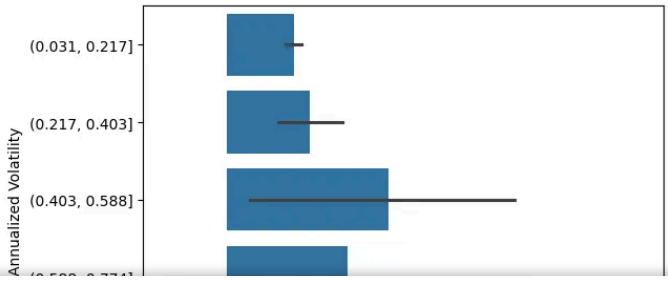
There are **fundamental differences** here that makes this comparison impossible: buy-the-dip assumes you have cash to allocate, while volatility targeting assumes t you are fully invested (calculated based on the current volatility level).

Although you could add to a position if it gets volatile (which usually means its price falling), that'd would be more of *"buy more when in drawdown"* technique. But that sounds pretty dangerous, kind of the opposite of a risk management strategy.

Type your email... Subscribe

The paradox

We looked at what the average returns were when the SP500 was in different drawdowns. Let's look at what they are when volatility is in a certain range [3]:



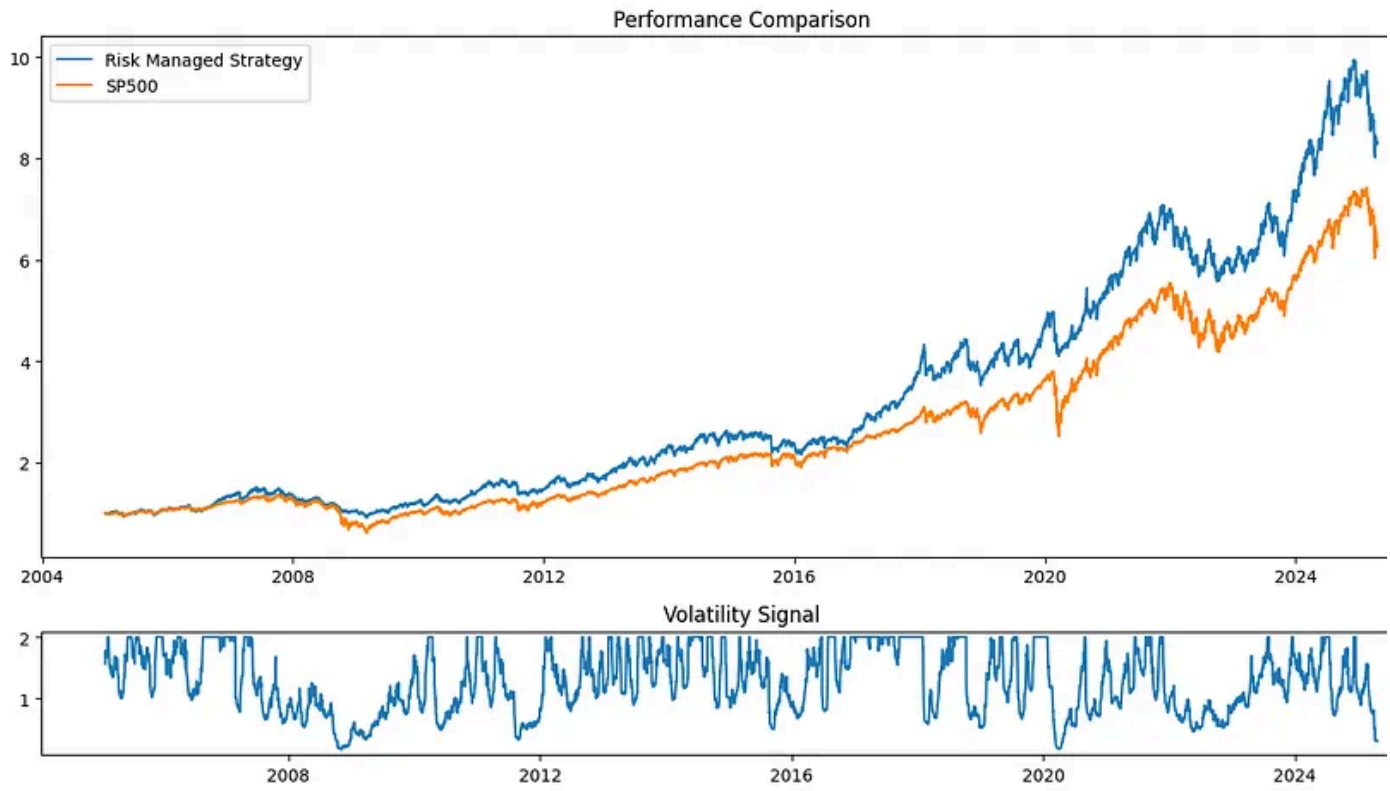
Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

plots.

When volatility was really high, we get higher average forward returns, than in the lo volatility regime - there's a *catch* of course. But that's the **opposite** of what we're looking for!

Let's look at the simplest possible backtest we could create, where we use volatility targeting with a 20 day rolling window on the SP500, rebalancing daily, with fixed 0 transaction costs:



You usually get 10-20% increase in risk-adjusted returns when you "blindly" [4] apply volatility targeting on an asset. [5]

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

mitigate its effect by averaging log returns instead of arithmetic returns, per Kris' explanation.

Try it yourself

We released a small python package (with MIT Licence) that helps you apply volatil targeting on any return series.

It's a collection of transparent, easy-to-audit functions, that we wrapped it into a package called `risklab` on pypi — it's super easy to use:

```
vol_signal = scale_to_target_volatility(
    target_volatility=0.16, # the annualized volatility target
    rolling_window=20,
    returns=sp500_data["Close"].pct_change(),
    upper_limit=2.0,
    lag=0,
    fill_initial_period_with_mean=False,
    annualization_period=252,
)
results = backtest_signal(
    signal=vol_signal,
    underlying=sp500_data["Close"].pct_change(),
    transaction_cost=0.001,
    lag=0,
)
```

Explore what asset classes volatility targeting works best for using `risklab` and th

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

Is this a silver bullet? What am I missing?

I personally always felt a bit cautious using only past volatility to manage risk. It basically means you're over-leveraged at times when volatility is very low, to compensate for reduced leverage when its high. Looks fine in a backtest, but it's a more uncomfortable when you are actually trading.

The missing piece is the realization that:

> ***not all periods of very low volatility are created equal.***

Unexpected events may hit when:

- Volume, liquidity is structurally low
- Implied volatility and the volatility risk premium is low
- Leverage is high

In those cases, you probably see massive moves, "de-leveraging" events where people have to reduce their exposure. Some may call it "Risk parity sell-of" or "Carı trade unwinds" - and their effect are probably **exaggerated** by the widespread usa of *volatility-based risk management itself*!

Last summer, implied & realized volatility were all at the lowest points in the last cou of years. VIX term structure was extremely flat, volumes were low (usual summer break vibes) — all of these were signs that a small sell-off could trigger a lot bigger one.

Can we use these and similar, asset-specific, or market-wide risk factors to reduce

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

As always, happy to connect on LinkedIn or X.

Type your email... Subscribe

This is part 1 of an in-depth investigation of how quantitative risk management cou help improve risk-adjusted returns. We'll cover the mechanics, the implementation details, benchmark the different "risk proxies", ensembling and a lot more!

---

1 Although there's definitely an artifact here: the rolling window calculation.

2 *These factor plots are misleading, as they're using equal bin size, and the number of observations in each bin are very skewed - the market is mainly in a low-volatile state.*

3 *These factor plots are misleading, as they're using equal bin size, and the number of observations in each bin are very skewed - the market is mainly in a low-volatile state.*

4 "Blindly" in this case does mean using a 20 day lookback window, coming from reasona priors, but there's always a chance that this is overfitting already.

5 Although it didn't perform particularly well in equities after 2008!

Subscribe to Unexpected Correlations by Unravel

Launched a year ago

Discover the predictive factors shaping market movements. In-depth reports, backtests and deploya code. By Unravel (unravel.finance): Market-netural, cross-sectional, multi-factor crypto portfolios. Joi Telegram for updates: t.me/unravel_finance

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

27 Likes • 2 Restacks

Next

Discussion about this post

Comments Restacks

Write a comment...

Tobi 1 May 2025

Liked by Mark Aron Szulyovszky

Great post, Mark! It really hit home, as I typically scale assets inversely to the difference betw their 6-month high and 6-month low, which meaningfully improves the risk/return profile of m strategies.

LIKE (1) REPLY

2 replies by Mark Aron Szulyovszky

Mark Aron Szulyovszky 22 Apr 2025

Author

Just started an in-depth series on quantitative risk management: exploring volatility targeting seemingly paradox phenomena and the blindposts associated with it.

LIKE REPLY

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.

© 2026 Unravel · Privacy · Terms · Collection notice
Substack is the home for great culture

Cookie Policy

We use cookies to improve your experience, for analytics, and for marketing. You can accept, reject, or manage your preferences. See our privacy policy.